

# EcoGuard Japan — Environmental Bulletin

**2026-06-02T09:34:32.118000+00:00 to  
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## ### Analysis of Air Quality Readings and Alerts

### #### General Observations:

#### 1. \*\*Air Quality Index (AQI)\*\*:

- The AQI values across major cities in Japan range from a low of 30 (Kyoto, Sapporo) to a high of 59 (Nagoya).
- Tokyo (AQI 36) and Kyoto (AQI 30) remain in the "Good" category, whereas Nagoya and Fukuoka (AQI 59 and 53, respectively) are on the verge of "Moderate".

#### 2. \*\*PM2.5 Levels\*\*:

- PM2.5 levels show significant variation. Nagoya presents the highest PM2.5 concentration of  $59.0 \mu\text{g}/\text{m}^3$ , which corresponds with its high AQI. Fukuoka and Osaka similarly have elevated PM2.5 levels ( $53.0 \mu\text{g}/\text{m}^3$  and  $42.0 \mu\text{g}/\text{m}^3$  respectively).
- Conversely, Hiroshima displays a notably low PM2.5 concentration at  $1.0 \mu\text{g}/\text{m}^3$ , which correlates with its AQI of 40, suggesting good air quality.

#### 3. \*\*Temperature and Humidity\*\*:

- Humidity ranges from a low of 71% (Osaka) to a high of 97% (Sendai). Elevated humidity levels were reported in Tokyo and Fukuoka, reaching 82% and 84%, respectively.
- Temperature readings vary slightly across cities with Fukuoka being the warmest at  $23.1^\circ\text{C}$  and Sapporo the coolest at  $19.0^\circ\text{C}$ .

### ### Alerts Overview:

- A notable number of alerts indicate humidity spikes in Tokyo, Fukuoka, and Sendai, all exceeding the threshold of 80%. This trend suggests adverse weather conditions that may influence air quality.

### ### Patterns and Hypotheses:

#### 1. \*\*Correlation Between Humidity and PM2.5 Levels\*\*:

- It can be hypothesized that elevated humidity is contributing to poor dispersion of air pollutants in cities with high PM2.5 levels (like Nagoya and Fukuoka). High humidity can lead to increased particulate matter and lower atmospheric mixing, causing elevated readings.

#### 2. \*\*Geographical Variability\*\*:

- Urban areas (Tokyo, Osaka, Nagoya) show higher PM2.5 levels compared to less densely populated areas like Kyoto and Hiroshima. This may indicate the impacts of urban traffic, industrial emissions, and localized pollution.

sources.

3. **\*\*Anomalous Readings in Hiroshima\*\***:

- Hiroshima's exceptionally low PM2.5 level could suggest effective mitigation strategies or environmental factors unique to the area (such as geographical location, regulation levels, etc.) that warrant further research.

4. **\*\*Potential for Extreme Weather Events\*\***:

- Given the high humidity levels in Sendai, an investigation into atmospheric conditions and their relation to increased air pollution levels during high humidity events may be warranted. Particularly, it would be prudent to consider whether these humidity spikes might be related to seasonal climatic phenomena.

**### Conclusion:**

The data presents an opportunity to further explore how humidity influences air quality across Japan. Coordinated efforts in monitoring can elucidate underlying causes for air pollution spikes, informing better policy decisions for public health and environmental management. Continuous observation and analysis will be critical in detecting trends related to climate change and urbanization's overall impact on air quality across Japan.